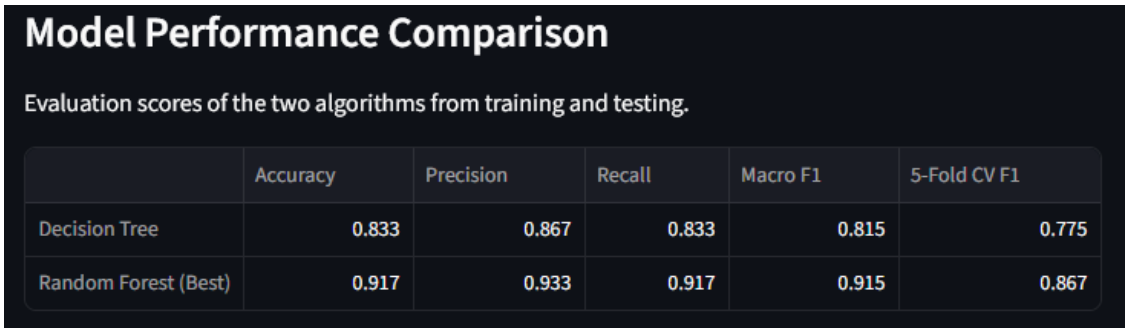
Facility Accessibility Classification

Model Comparison



Model Performance Comparison

Evaluation scores of the two algorithms from training and testing.

	Accuracy	Precision	Recall	Macro F1	5-Fold CV F1
Decision Tree	0.833	0.867	0.833	0.815	0.775
Random Forest (Best)	0.917	0.933	0.917	0.915	0.867

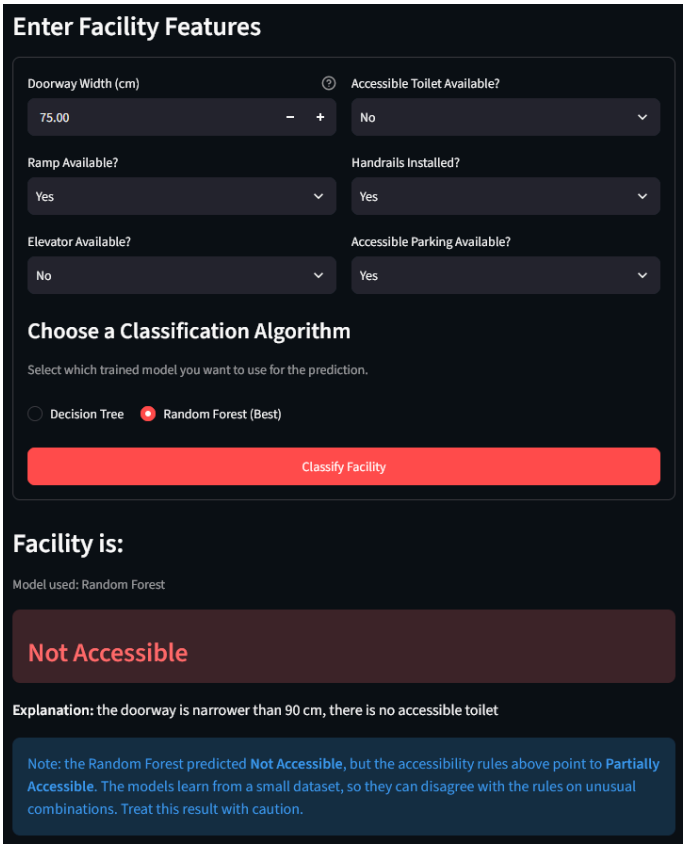
As shown in the table, the Random Forest outperformed the Decision Tree on every metric, with an accuracy of 0.917 compared to 0.833 and a 5-fold cross validation F1 score of 0.867 compared to 0.775. Because it had the higher cross validation F1 by a clear margin, the Random Forest was selected as the best performing model for the Streamlit application.

Interpretation of Results

Both models classified every Accessible and Not Accessible facility in the test set correctly. The difference came from the Partially Accessible category, where the Decision Tree misclassified two of four facilities and the Random Forest misclassified one. This is the hardest category because it lies between the two extremes, so a facility only needs to miss one requirement, such as a missing toilet or a doorway just under 90 cm, to cross a boundary. The Random Forest likely handled these borderline cases better because it combines the votes of 100 trees, which reduces the overfitting a single Decision Tree is prone to. Feature importance also showed that doorway width was the most influential input for both models, which is consistent with the 75 cm and 90 cm thresholds used to label the dataset.

Example Prediction

Results below show a borderline case entered into the Streamlit app: a 75 cm doorway with a ramp, handrails and accessible parking, but no elevator and no accessible toilet. The Random Forest classified this facility as Not Accessible, while the labeling rules used for the dataset would classify it as Partially Accessible, since the doorway is not narrower than 75 cm and a ramp provides step-free access. The app flags this to the user.



Limitations

- **Small dataset -** The dataset has only 60 records and the test set only 12, so a single misclassified facility noticeably changes the accuracy score. The difference between the models may therefore partly reflect chance.
- **Rule-based labels -** The labels follow rules defined by the group (step-free access and 75 cm / 90 cm doorway thresholds), so the models learn those rules rather than real-world accessibility judgments.
- **Limited features -** Only six features were used. Factors such as ramp slope, signage, door type, and floor surface were not considered.
- **No hyperparameter tuning -** Both models used default settings with a fixed random seed.
- **Borderline cases -** As the Example Prediction shows, the models can disagree with the rules near the thresholds, so predictions for unusual combinations should be treated with caution.

Overall, the Random Forest is the more reliable of the two algorithms for this task, but a larger and more varied dataset would be needed before the results could be trusted beyond this project.

Critical-thinking questions

1. Which input features should have the strongest influence on the assigned category?
    - Doorway width and Ramp availability should have the strongest influence, because they can decide whether a person can use the toilet at all.

2. What should the system do when the available evidence is incomplete or contradictory?
    - The system should not give an answer. It should ask for missing values as the app already does here like for doorway width, and give its output and warn the user when the system or model finds the toilet not within accessibility rules.

3. What could happen if the system produces an incorrect result for facility accessibility classification?
    - If the system labels an inaccessible facility as Accessible, a person with a mobility impairment could travel there and find they cannot enter or use it, which wastes their time. If it then labels an accessible facility as Not Accessible, people may avoid the facility they could have used. And wrong results can also make users lose trust in the system or application because of inconvenience.